

AgentMira

Engineering Case Study

The Buyer Lead Intake Agent

About this case study

Congratulations on making it to this round. We were impressed by your background and would like to evaluate how you approach real engineering problems of the kind we solve at AgentMira every day.

This case study is intentionally open-ended. There is no single "correct" answer. We are evaluating how you think — how you approach an ambiguous problem, how you design a solution, and how you explain your reasoning. Take your best shot at the objective described below.

Context

AgentMira is an AI-native real estate platform built for licensed realtors. Realtors are real estate agents that help in sale and purchase of homes for the end consumers. We help realtors do their best work by giving them AI teammates that handle the time-consuming parts of their job — processing inbound leads, researching properties, preparing buyer briefs, and coordinating with clients.

A real estate agent receives many buyer leads every week through their website, referrals, and inbound calls. A buyer lead is a person who has a need to buy a home and has yet not decided which realtor they would like to work with. Each lead is a free-text message describing what someone is looking for.

Processing these leads well — understanding what the buyer actually wants, finding matching properties, and giving the realtor enough context to follow up effectively — is one of the most valuable things our platform does. Lead conversion directly correlates with the money realtor can make as more leads eventually means more transactions through them and hence more money they can earn.

Your task is to build a version of this capability.

The objective

Build a **Buyer Lead Intake Agent** that takes an incoming buyer inquiry and produces a **Lead Brief** that helps the realtor respond effectively.

The agent should:

- Process the incoming buyer inquiry.
- Match the buyer's needs against the MLS dataset we provide.
- Produce a structured Lead Brief that the realtor can act on quickly.

The Lead Brief is the document the realtor reads before reaching out to the buyer. It should give them everything they need to have an informed first conversation.

The data we are providing

You will receive two files alongside this brief:

File	What it contains
miami_mls_listings.csv	Approximately 300 active and recent property listings from the greater Miami area. Includes address, neighborhood, price, bedrooms, bathrooms, square footage, features, status, dates, and other listing metadata.
sample_buyer_inquiries.json	Twelve sample buyer leads received through various channels (website forms, referrals). Each contains the buyer's contact info and free-text message describing what they are looking for.

This is the same shape of data we work with in production, though obviously a smaller subset. Treat it as real — meaning: real data has the rough edges that real data has.

What the Lead Brief should contain

We are leaving the exact structure of the Lead Brief up to you — part of what we are evaluating is your judgment about what a realtor actually needs to act on a lead. At minimum, we would expect each brief to include:

- A clear summary of what the buyer is looking for, extracted from their message.
- Recommended properties from the MLS dataset that match the buyer's needs, with reasoning for why each is a match.
- Anything the realtor should be aware of before reaching out to this buyer — context, concerns, or judgment calls that need a human.
- A suggested next action for the realtor.

The format is up to you (JSON, Markdown, structured text — whatever you think communicates best to a busy realtor reviewing it on a phone or laptop).

What to submit

We need three things from you:

- 1. Source code** — Your full implementation, either as a GitHub repository (preferred) or a zip file. Include a README with clear instructions to run it. Use whatever language, framework, and libraries you want.
- 2. The 12 Lead Briefs** — Generated by your agent for each of the 12 sample inquiries. Submit them in whatever format your agent produces.

3. A written explanation (2–3 pages) — This is the most important part of your submission. It should cover:

- Your overall approach and design decisions — what you built, why you built it that way, and what tradeoffs you made.
- Your walkthrough of the 12 Lead Briefs — for each inquiry, briefly explain what your agent decided to do and why. If anything in an inquiry or in the data made you pause, tell us about it.
- How you used AI coding tools during the build (which tools, where they helped, where you had to override or correct them).
- What you would do differently or build next if you had more time.

Optional but appreciated: a short (under 5 minutes) Loom or screen recording walking us through your solution.

Constraints

Timeline: You have **two calendar days** from when you receive this brief. Submit by end of day on day two.

AI coding tools: You are **free** to use AI coding tools (Claude Code, Cursor, GitHub Copilot, Codex, or any equivalent).

Tech stack: Your choice entirely. Python, TypeScript, Go — whatever you are most productive in. Same for frameworks: use what makes sense, not what sounds impressive.

LLM provider: Your choice. Use whatever you have access to (Claude, OpenAI, Gemini, Llama, etc.). If cost is a barrier, mention it in your writeup — we are not testing whether you can afford API calls.

No external human help: Other than AI coding tools, this should be your own work. Don't share the brief or your code with others. Honesty and Integrity is one of the core values that we have in the organization and that is best displayed when no one is watching you.

How we will evaluate your submission

We are looking at five things, in rough order of weight:

Dimension	What we are looking for
Agent design	Does your solution actually demonstrate agent-like reasoning, or is it a thin wrapper around a single LLM call? How are tools, decisions, and intermediate steps structured?
Quality of outputs	Would a real realtor find your Lead Briefs useful? Are they accurate, well-organized, and actionable across the variety of inquiries you receive?
Engineering judgment	How well did you handle the realities of the problem? Did you notice things that needed attention? Did you make sensible decisions where the brief did not spell things out?
Explanation	How clearly do you reason about what you built and why? Your writeup is where we learn how you think — invest in it.
Code quality	Is the code clean, organized, and maintainable? We are not looking for perfection in two days, but we are looking for someone we'd want to work alongside.

A note before you start

Take assumptions if something is not clear, or you feel it is incomplete. Real data and real world situations will not be perfect. We are far more interested in how you think than in any specific output. Build something you would want to use if you were the realtor on the receiving end. *Good luck. We are looking forward to your submission.*